PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	The association between social supports and depression among diabetes mellitus patients in Ethiopia: a systematic review and meta-analysis
AUTHORS	Diress, Gedefaw; Endalifer, Melese; Addisu, Amanuel; Mengist, Belayneh

VERSION 1 – REVIEW

REVIEWER	Teresa Neeman Australian National University, Biological Data Science Institute
REVIEW RETURNED	01-Mar-2022

GENERAL COMMENTS	Overall: This paper reads much better now that causation language has been replaced by associations. The paper would benefit by including a table similar to a Summary of Findings table seen in Cochrane reviews. The key information in this table would include general levels of depression in this cohort both with and without social support. This would help the target audience understand the relevance of an estimated OR = 2. If depression levels vary widely between regions, then such a table could include typical levels for different types of regions. This meta-analysis is unusual in that it includes a single outcome. Secondary outcomes (whilst not essential) could add more robustness to the conclusions. I felt that the paper could be stronger if some of the sensitivity analyses were presented in an appendix rather than in the main body of the paper. Discussion: There were some incorrect statistical claims made in the discussion. These should be changed. They are: Line 265: Subgroup analysis shows that the effect size was larger in studies with participants' mean age ≤45 years compared to mean age>45 years. Comment: The subgroup analysis does not show that the effect sizes are different. The ESTIMATES of the effect sizes are different, but the very wide confidence intervals precludes one from claiming that there are differences in the true effect sizes. Line 267: Similarly, social support is significantly associated with depression in studies with a sample size of ≥400. But, there is no association in studies with a sample size of <400.

	Comment: The authors indicate by this comment that studies with >= 400 participants show a greater social support effect compared with studies with <400 participants. This reasoning is fallacious. P>0.05 in one set of studies could merely indicate that there were fewer participants included (ie insufficient power). Note that any claim about comparing effect sizes between subgroups needs be supported statistically. Line 269: In the sensitivity analysis, no significant change in heterogeneity was reported when excluding any individual study suggesting homogeneity of the pooled effect estimates. Comment: This is incorrect. The I^2 is a measure of heterogeneity between studies. It was 71% and is driven mostly by two studies that were different to the others. It would be better to acknowledge the sources of heterogeneity and provide a discussion around why those 2 studies might be different.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1

Comment 1: *This paper reads much better now that causation language has been replaced by associations.*

Response1: Thank you very much.

Comment 2: *The paper would benefit by including a table similar to a Summary of Findings table seen in Cochrane reviews. The key information in this table would include general levels of depression in this cohort both with and without social support. This would help the target audience understand the relevance of an estimated OR = 2.*

Response2: Thank you very much for your valuable comment. However, in this study, we included articles only that reported the estimated RR/OR with their 95% CI or articles that reported necessary row data used to calculate OR or RR. In this meta-analysis, out of seven studies, only four studies reported the proportion of depression by the level of social support. So, we couldn't present the proportion of depression among good social support vs poor/no social support for all included studies. That is why we missed the proportion of depression by social support in the manuscript. If there is another option, we are ready to incorporate the summary finding.

Comment 3: *If depression levels vary widely between regions, then such a table could include typical levels for different types of regions.*

Response 3: Sorry, in this study, we stated only the variation in effect size (odds ratio) across regions, not on depression level. The magnitude of depression was not assessed in the current study. Therefore, we didn't perform a subgroup analysis for the level of depression by region because the main objective of this meta-analysis was only to assess the association between social support and depression. As a means of investigating heterogeneity, we performed a subgroup analysis by geographical location(region) and the result showed that a significant association was only found among studies done in regions other than Amhara and Addis Ababa (OR= 2.07, 95% CI = 1.40–3.08), but not in Amhara and Addis Ababa.

Comment 4*: This meta-analysis is unusual in that it includes a single outcome. Secondary outcomes (whilst not essential) could add more robustness to the conclusions*

Response 4: we agree with the comment but we extracted data only for a single question (does social support associated with depression among patients with diabetes?). If you suggest another important outcome, we are ready to incorporate other important outcomes.

Comment 4: *I felt that the paper could be stronger in some of the sensitivity analyses were presented in an appendix rather than in the main body of the paper****.***

Response 4:- Thank you for your suggestion. Based on your comment, we presented the sensitivity analyses section in the appendix.

Discussion

There were some incorrect statistical claims made in the discussion. These should be changed. They are:

Line 265: Subgroup analysis shows that the effect size was larger in studies with participants' mean age ≤45 years compared to mean age>45 years.

Comment 5: *The subgroup analysis does not show that the effect sizes are different. The ESTIMATES of the effect sizes are different, but the very wide confidence intervals precludes one from claiming that there are differences in the true effect sizes.*

Response 5: Yes you are right. Thank you for this valuable comment. It is corrected in the revised manuscript.

Comment 6 *Line 267: Similarly, social support is significantly associated with depression in studies with a sample size of ≥400. But, there is no association in studies with a sample size of <400.*

Comment: The authors indicate by this comment that studies with >= 400 participants show a greater social support effect compared with studies with <400 participants. This reasoning is fallacious. P>0.05 in one set of studies could merely indicate that there were fewer participants included (ie insufficient power).

Note that any claim about comparing effect sizes between subgroups needs be supported statistically.

Response 6: Thank you. corrected

Comment 7: *Line 269: In the sensitivity analysis, no significant change in heterogeneity was reported when excluding any individual study suggesting homogeneity of the pooled effect estimates.*

Comment: This is incorrect. The I^2 is a measure of heterogeneity between studies. It was 71% and is driven mostly by two studies that were different to the others. It would be better to acknowledge the sources of heterogeneity and provide a discussion around why those 2 studies might be different.

Response 7: we agree with the comment. We modified the sentence as follow: To detect the studies which influence the overall estimate of our meta-analysis, we performed sensitivity analysis. The results of the sensitivity analysis, which was done by excluding studies one by one, revealed no significant change in the pooled estimate when excluding any individual study.

VERSION 2 – REVIEW

REVIEWER	Teresa Neeman Australian National University, Biological Data Science Institute
REVIEW RETURNED	06-Apr-2022

GENERAL COMMENTS	A few minor comments: Reporting subgroup analyses: Authors should not focus on statistical significance or non-significance of the results. A non-significant p-value simply means "insufficient evidence of an effect" which can be the result of a true difference but insufficient sample size. For me, the purpose of a subset analysis is to assess the consistency of the treatment effect across subsets. When describing the subgroup analysis, the authors should report OR and 95% CI and comment on the absence of inconsistency in effect sizes between subsets or between the subset and the overall sample. In other words, they can report in a similar way to how they report their sensitivity analyses.

	I would like to see depression rates (x/n , %) for each study. Usually, one can find this in the forest plot (see Cochrane reviews). In my last review, I suggested a Summary of Findings table in order to see an indicative depression rate for non-supported diabetic patients, and how social support might modify this. But I'd be happy with a revised forest plot that includes depression rates in each cohort, as well as the OR for each study. Check number of digits in each Table and ensure there is consistency. I think reporting CI to 0.xxx is too many digits. I don't need to review this again as these are proposed changes can be easily implemented.